Assessing marginal water values in multipurpose multireservoir systems via stochastic programming

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[1] The International Conference on Water and the Environment held in Dublin in 1992 emphasized the need to consider water as an economic good. Since water markets are usually absent or ineffective, the value of water cannot be directly derived from market activities but must rather be assessed through shadow prices. Economists have developed various valuation techniques to determine the economic value of water, especially to handle allocation issues involving environmental water uses. Most of the nonmarket valuation studies reported in the literature focus on long-run policy problems, such as permanent (re)allocations of water, and assume that the water availability is given. When dealing with short-run allocation problems, water managers are facing complex spatial and temporal trade-offs and must therefore be able to track site and time changes in water values across different hydrologic conditions, especially in arid and semiarid areas where the availability of water is a limiting and stochastic factor. This paper presents a stochastic programming approach for assessing the statistical distribution of marginal water values in multipurpose multireservoir systems where hydropower generation and irrigation crop production are the main economic activities depending on water. In the absence of a water market, the Lagrange multipliers correspond to shadow prices, and the marginal water values are the Lagrange multipliers associated with the mass balance equations of the reservoirs. The methodology is illustrated with a cascade of hydroelectric-irrigation reservoirs in the Euphrates river basin in Turkey and Syria.

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1. Introduction

[2] Integrated Water Resources Management (IWRM) is the current paradigm for planning and managing our water resources. IWRM is now advocated by international and Northern bilateral agencies and is being considered by many water professionals as applicable worldwide even though some authors are challenging this idea [*Biswas*, 2004]. IWRM relies on four principles which have been debated during the Dublin conference on Water and the Environment in 1992. One of those principles states that water should be considered as an economic good and that adequate measures should be adopted to guarantee equity. How this economic principle should be interpreted is still debated among water professionals. So is its applicability. The objective of this paper is neither to further discuss this principle nor to expose another point of view. Interested readers should refer, for example, to *Briscoe* [1997], *Savenije and van der Zaag* [2002] and *Davis* [2007] for an overview of the different schools of thought.

[3] This paper rather addresses the issue of assessing the marginal value of water and its statistical distribution at various locations both in space and time. The marginal value of water gives the contribution of an additional unit of water to whatever public or private objective is under consideration. Assessing this value is not an easy task because water is a fugitive and stochastically supplied resource, certain water uses may generate externalities, and markets are more than often non applicable. According to *Rogers et al.* [2002] the value of water is made up of several components some of which are not easily monetized: (1) the marginal value of product (willingness to pay), (2) benefits from return flows, (3) multiplier effects from indirect uses, (4) the benefits for meeting societal objectives, and finally, (5) the intrinsic value which consists of cultural, aesthetic, and merit values. Together these five components constitute the full value of water which should be compared to the full cost of water when allocating water between different users. When water can be considered as a variable input in production, the value of water can be measured by the value marginal product; that is the value of an additional quantity of product due to the use of an additional unit of water.

[4] Marginal water values can be used in river basin management to signal water scarcity, to prioritize zones where soil and water conservation measures must be implemented, to increase the productivity of water by (re)allocating it to more productive uses. Since markets are usually absent or ineffective, allocation decisions can seldom rely on market prices. Instead, accounting or shadow prices reflecting the value of water must be assessed. To achieve this, economists

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have developed and implemented various nonmarket valuation techniques to water resources management problems [*Loomis*, 2000], especially in the field of water quality such as stream and wetland restoration [see, e.g., *Collins et al.*, 2005; *Woodward and Wui*, 2001]. Despite the extensive work being carried out in environmental valuation, there is no consensus among economists on the meaning and use of environmental valuation in policy [*Shabman and Stephenson*, 2000]. A review of different economic valuation techniques applied to water resources management problems is given by *Birol et al.* [2006] and by *Young* [2005].

[5] Most of the nonmarket valuation studies reported in the literature focus on single water use, long-run policy problems, and assume that the water availability is given. *Gibbons* [1986] provides examples of average water values for different uses in the US. A recent review of water values for different uses in Scotland is given by *Moran and Dann* [2007]. *Rogers et al.* [1998] present separate costs and values for irrigated agriculture, urban and industrial uses for three case studies in Thailand and in India. Average economic values of various wetland's goods and services are listed by *Schuyt and Brander* [2004]. To the best of our knowledge, there is no study which tracks site and time changes in values across different hydrologic conditions. But it is precisely this type of study that is of relevance for water resources managers who are facing complex spatial and temporal trade-off when allocating water in a stochastic environment. The objective of this paper is to present a stochastic approach based on systems analysis to assess marginal water values and their statistical distributions at various points in space and time.

[6] There are essentially two modeling approaches in water resources systems analysis to derive marginal water values: simulation versus mathematical programming. Both approaches require the coupling of a hydrological model with an economic model. With the simulation approach, the marginal water values can only be derived after analyzing the results of a large number of runs associated with small perturbations in water availability. This approach has been adopted by *Seyam et al.* [2002]. The main advantage is that it can handle nonlinear relationship and can be fairly detailed. The disadvantages come from the fact that it is computationally cumbersome as the number of simulations is quite large.

[7] Generally speaking, the mathematical programming approach consists in optimizing an economic objective function subject to constraints representing, among other things, the governing equations of the hydrological processes. With this approach, the marginal water values correspond to the Lagrange multipliers associated with the mass balance equations which are all available at the optimal solution. The Lagrange multipliers give the change in the objective function due to a small perturbation in the constraint right-hand side. These multipliers, also called shadow prices, would correspond to market prices if water were being traded on a market. According to *Young* [2005], mathematical programming is one of the deductive methods for nonmarket valuation of water. The mathematical programming approach has been adopted, for example, by *Rosegrant et al.* [2000] and *Cai et al.* [2003] to analyze river basins characterized by an important agricultural sector. A similar approach is adopted by *Ringler et al.* [2004] to analyze development scenarios in the Mekong river basin. *Ringler and Cai* [2006] investigate different water allocation strategies and trade-offs in the Mekong River Basin after incorporating the water values for fisheries and wetlands into an integrated economic-hydrologic model. *Wang et al.* [2007] present various models based on mathematical programming to allocate water between competing users. *Watkins and Moser* [2006] determine optimal operating rules for two reservoirs of the Panama Canal System using a deterministic optimization model. Optimization models can also be developed for groundwater resources [*Shamir et al.*, 1984; *Das and Datta*, 2001].

[8] Most of the integrated economic-hydrologic models reported in the literature do not explicitly consider the hydrologic uncertainty and are therefore likely to overestimate the performance of the water resources system. However, the advantages of adopting stochastic formulations when the problem has no certainty equivalent have been discussed for reservoir operation problems [*Philbrick and Kitanidis*, 1999]. More specifically, the probabilistic description of the hydrologic uncertainty found in stochastic models implies that there is no need for certainty equivalence as the resulting policies provide a hedging strategy against extreme hydrological events such as floods and droughts. Since allocation policies and water values are intertwined, stochastic programming should be used if one wants to avoid errors associated with deterministic formulations. In the work by *Hollinshead and Lund* [2006], for example, a stochastic three-stage linear program optimizes environmental water purchases for different hydrologic scenarios using physical and financial assets. The use of optimization models is also widespread in the hydropower industry where the marginal value of water is a key factor for scheduling reservoir releases and thus hydropower generation [*Wallace and Fleten*, 2003]. The basic idea is that a release is optimal when the immediate and expected future water values are equal. Examples of optimization models in hydrodominated systems are given by *Pereira and Pinto* [1991], *Tejada-Guibert et al.* [1995], *Scott and Read* [1996], and *Kim et al.* [2007].

[9] This paper presents a stochastic programming model that can be used to assess marginal water values in multipurpose multireservoir systems taking into account the hydrologic uncertainty. The model, called stochastic dual dynamic programming (SDDP), is implemented on a cascade of 7 multipurpose reservoirs in the Euphrates river basin. As its name indicates, SDDP is an extension of SDP that removes the curse of dimensionality found in discrete SDP. This paper analyzes the marginal water values at various points (reservoirs) and time (month) for a system where the benefits are essentially derived from energy and agricultural sales. The paper also extends the SDDP model described by *Tilmant and Kelman* [2007] by including irrigation benefits directly in the objective function and by taking into account the topology of irrigation return flows. To handle the issues of nonadditivity and dependence of irrigation benefits on previous allocation decisions, additional state variables representing "dummy" reservoirs of accumulated water diverted to the crops are included in the state vector of the SDDP algorithm. At the end of the irrigation season, we assume that agricultural benefits are proportional to the accumulated water that has been diverted to the crops since their planting dates. The ability of the SDDP algorithm to cope with a large state space has made

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the inclusion of these additional states variables in the state vector possible.

[10] The type of water value to be assessed depends on the policy problem at hand. In this study we analyze midterm allocation policies that can capture the stochasticity of the hydrologic input and, therefore, provide hedging strategies against the hydrological risk. Since the monthly time step is fairly common in midterm (strategic) studies of hydropower systems [*Labadie*, 2004], the SDDP model identifies monthly release and irrigation withdrawal policies that maximize the net benefits associated with the operation of the system. Consequently, the model calculates short-run marginal water values; the fixed costs (e.g., dams, canals, hydropower plants) having zero opportunity costs are thus ignored. In addition, the model provides estimates of the marginal value of product and the benefits from return flows as existence values and the benefits from indirect uses are ignored.

[11] This paper is organized as follows. Section 2 reviews multistage decision-making problems. Stochastic dynamic programming (SDP) and stochastic dual dynamic programming (SDDP) are explained next in sections 3 and 4, respectively. Then, section 5 explains how irrigation benefits are incorporated into the objective function. The Euphrates river in Turkey and Syria and the corresponding SDDP model are described sections 6 and 7. Finally, simulation results focusing on water values are presented and analyzed in section 8, while conclusions and recommendations for future works are given in section 9.

2. Multistage Decision-Making Problems

[12] The optimal operation of a multireservoir system can be considered as a multistage decision-making problem: a sequence of T release decisions r_t must be determined so that the overall expected benefit Z is maximized:

$$Z = E\left[\sum_{t=1}^{T} f_t(u_t, r_t, q_t) + v(u_{T+1})\right] \tag{1}$$

where f_t is the immediate benefit of system operation during period t, q_t is the vector of inflows to the reservoirs during period t, u_t is the state vector describing the system's status at the beginning of period t, r_t is the vector of release decisions during period t, and $v(u_{T+1})$ is a terminal value function. In many reservoir operation problems the state vector includes a storage state variable s_t and a hydrologic state variable such as the previous flow q_{t-1} [*Tejada-Guibert et al.*, 1995].

[13] In a hydropower system, a typical immediate benefit function will include the net benefits from hydropower as well as penalty costs for not meeting target water demands and/or violating operating constraints. For example, one may have

$$f_t(s_t, r_t, q_t) = \tau_t \sum_j (\pi_t^h(j) - \theta(j))e(j)r_t(j)h_t(j) - \xi_t' x_t \tag{2}$$

where π_t^h is the energy price at time t for the jth hydropower plant [\$/MWh]; $\theta(j)$ is the operation and maintenance cost of the jth hydropower plant [\$/MWh]; τ_t is the number of hours in period t; $e(j)$ is the efficiency function associated with the jth hydropower plant, which depends on the net head $h_t(j)$ and the release $r_t(j)$ [MW/m.m^{3}s^{-1}]; ξ is the vector of penalty coefficients [\$/unit deficit or surplus]; x_t is the vector of deficits or surpluses [unit deficit or surplus]. Irrigation rationing and/or the inability to meet minimum flow requirements are examples of deficits. Surpluses like unsafe storage levels for flood control purposes and/or spillages losses may also be penalized.

3. Stochastic Dynamic Programming

[14] In stochastic dynamic programming (SDP) release decisions are made to maximize current benefits plus the expected benefits from future operation, which are represented by the recursively calculated benefit-to-go function F_{t+1}. Considering the reservoir operation problem as periodic and infinite-horizon the long-term optimal operating policy can be found by the recursive solution of the functional SDP equation

$$F_t(s_t, q_{t-1}) = \underset{q_t|q_{t-1}}{E}\left[\max\{f_t(s_t, q_t, r_t) + F_{t+1}(s_{t+1}, q_t)\}\right] \tag{3}$$

subject to

$$s_{t+1} - C_R(r_t + l_t) = s_t + q_t - e_t(s_t) - i_t \tag{4}$$

$$\underline{s}_{t+1} \le s_{t+1} \le \overline{s}_{t+1} \tag{5}$$

$$\underline{r}_t \le r_t \le \overline{r}_t \tag{6}$$

where l_t is the vector of spills; C_R is the system connectivity matrix ($C_R(j, k) = 1(-1)$ when reservoir j receives(releases) water from(to) reservoir k); e_t is the vector of evaporation losses; i_t is the vector of irrigation water withdrawals; $\underline{s}$ and $\overline{s}$ are vectors with the minimum and maximum storage volumes, respectively; $\underline{r}$ and $\overline{r}$ are vectors with the minimum and maximum releases, respectively.

[15] The principle of dynamic programming is illustrated in Figure 1 where we can see that the optimal solution is achieved when the sum of the immediate (f_t) and benefit-to-go functions (F_{t+1}) is maximum. As the volume of water left in storage at the end of stage t increases, the benefit-to-go function (F_{t+1}) increases as more water is available for future uses. Conversely, the immediate benefits decrease as less water is being used during the current stage t. The derivatives of the immediate and benefit-to-go functions give the immediate and future water values, respectively, which must be identical at the optimal solution.

[16] The recursive equation (3) is carried out until the change in the benefit-to-go function from one iteration to the next becomes nearly constant for each point of the discrete state space domain [*Loucks and van Beek*, 2005]. The resulting steady state release policy and benefit-to-go functions constitute the sets of solutions that can be used by reservoir operators to derive an optimal release policy. In practice, to solve the SDP model (3), the hydrologic state and the storage variables must be discretized into N_q and N_s intervals, respectively, each represented by a characteristic value [*Yakowitz*, 1982]. The continuous domain (s_t, q_{t-1}) is thus replaced by a grid so that an approximate solution of (3) can be developed by evaluating (3) at the grid points only.

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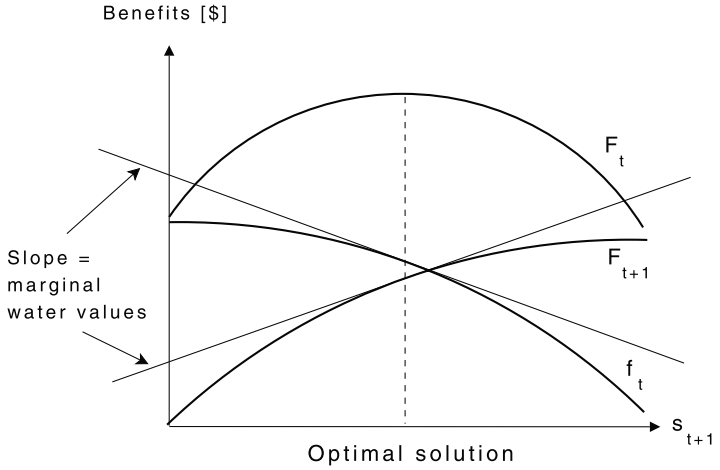


Figure 1. Immediate and future benefits functions.

The drawback of this discrete approach comes from the fact that the computational effort W required to solve the SDP model increases exponentially with the number of reservoirs $J(W \propto (N_s N_q)^J)$, making it unsuitable to handle systems involving more than 3–4 reservoirs.

4. Stochastic Dual Dynamic Programming

[17] Stochastic Dual Dynamic Programming (SDDP) is an extension of SDP that removes the computational burden found in discrete SDP. The SDDP algorithm was first developed by *Pereira* [1989] and *Pereira and Pinto* [1991] to dispatch the power plants of the Brazilian vast hydropower system. This algorithm is now used in several hydro-dominated countries such as Norway and New Zealand [*Scott and Read*, 1996; *Mo et al.*, 2001; *Kristiansen*, 2004]. SDDP removes the computational burden that characterizes SDP by constructing a piecewise linear approximation of F_{t+1} through sampling and decomposition. For each sampled values of the state variables a hyperplane is constructed and provides an outer approximation of the benefit-to-go function F_{t+1}. These hyperplanes are equivalent to Benders cuts in Benders decomposition [*Kall and Wallace*, 1994]. Intuitively, the computational effort should be reduced since the value of F_{t+1} can now be derived by extrapolation instead of interpolation as in SDP. In other words, a limited number of values for the state variables (points) are now sufficient to provide an approximation of F_{t+1} as illustrated in Figure 2.

[18] To further save computation time the SDDP algorithm first starts with a small number of points and then recursively constructs the cuts corresponding to these points starting at the last stage T and moving backward until the first stage t_0. The cuts generated during this backward

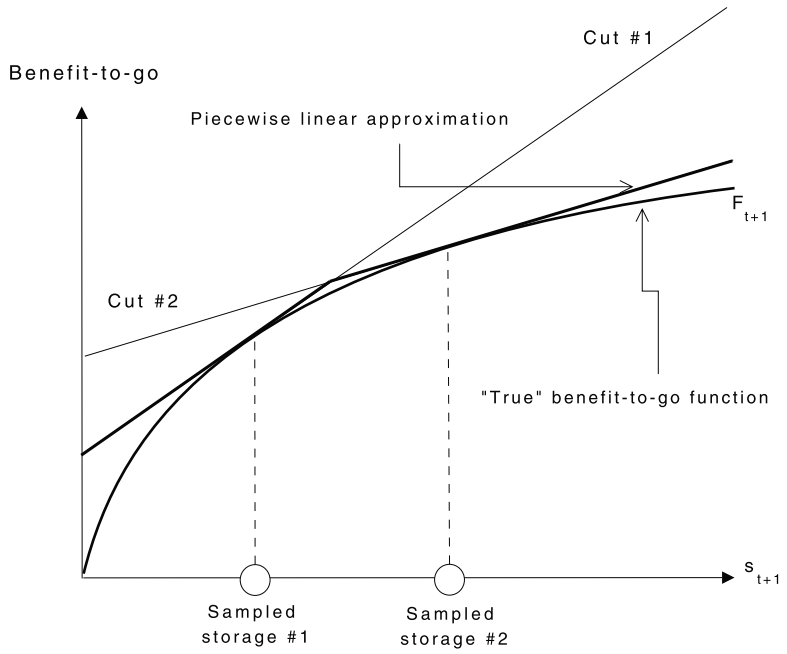


Figure 2. Piecewise linear approximation of F_{t+1}.

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optimization phase are then evaluated by checking whether the approximation they provide is statistically acceptable or not. This is done by simulating forward the system with (1) the cuts generated in the previous backward optimization phases and (2) several hydrologic sequences, which can be historical and/or synthetically generated. If the approximation is not acceptable, then a new iteration starts and a new backward optimization phase is implemented with a larger sample which now also contains the points the last simulation went through.

[19] Let φ_{t+1}^{l}, β_{t+1}^{l} and γ_{t+1}^{l} be the parameters of the lth cut approximating F_{t+1}. As explained by *Tilmant and Kelman* [2007], the values of these parameters have been determined previously, at stage $t + 1$, from the primal and dual information available after having solved the one-stage optimization problem (3) at that stage. As a matter of fact, in SDDP, the one-stage optimization problem (3) must be a convex program, such as a linear program (LP), so that the Kuhn-Tucker conditions for optimality are necessary and sufficient. In particular, the gradient of the objective function is equal to the linear combination of the gradients of the binding constraints and their corresponding dual information. It is therefore possible to evaluate the derivatives of the objective function F_{t+1} with respect to the state variables (s_{t+1}, q_t) from the derivatives of the binding constraints and the dual information, which are both available at the optimal solution at stage $t + 1$. This analytical determination of the gradients of the functional equation F_{t+1} will be explained later. We must first pose an important assumption regarding the structure of the immediate benefit function.

[20] Since hydropower is a nonlinear function of the head (storage) and release variables, it cannot be used as such in the immediate benefit function of SDDP. One way to remove this source of nonconvexity is to assume that the production of hydroelectricity is dominated by the release term and not by the head (or storage) term. This assumption is valid as long as the difference between the maximum and minimum heads is small compared to the maximum head. If the system is strongly nonlinear, alternative methods such as Neuro-DP can be employed to solve the multireservoir operation problem. In Neuro-DP, the one-stage optimization problem no longer needs to be convex and the benefit-to-go function is approximated by an artificial neural network which must be trained prior to solving the optimization problem [*Castelletti et al.*, 2007]. Another promising option is to use the Q-Learning method of reinforcement learning to derive the optimal operating policies through a forward looking depth-first procedure that alleviates the curse of dimensionality found in discrete SDP [*Lee and Labadie*, 2007]. However, since both Neuro-DP and reinforcement learning were implemented on rather small systems (with 3 and 2 reservoirs, respectively) it is still unclear whether they would be able to deal with a larger system as the one used to illustrate the SDDP approach.

[21] With this linear assumption the one-stage optimization problem becomes a LP problem and the decomposition scheme can be implemented. Mathematically, the immediate benefit function (2) must be rewritten as

$$f_t(s_t, r_t, q_t) = \tau_t \sum_j (\pi_t^h(j) - \theta(j)) c^h(j) r_t(j) - \xi_t^{\cdot} x_t \qquad (7)$$

where $c^h(j)$ is the production coefficient associated with the jth hydropower plant (MW/m^3s^{-1}).

[22] With the above immediate benefit function and using L cuts to approximate F_{t+1}, the one-stage optimization problem (3) becomes

$$F_t(s_t, q_{t-1}) = \max\{f_t(s_t, q_t, r_t) + F_{t+1}\} \qquad (8)$$

subject to

$$s_{t+1} - C(r_t + l_t) = s_t + q_t - e_t(s_t) - i_t \qquad (9)$$

$$\underline{s}_{t+1} \le s_{t+1} \le \overline{s}_{t+1} \qquad (10)$$

$$\underline{r}_t \le r_t \le \overline{r}_t \qquad (11)$$

$$f_t(s_t, r_t, q_t) = \tau_t \sum_j (\pi_t^h(j) - \theta(j)) c^h(j) r_t(j) - \xi_t^{\cdot} x_t \qquad (12)$$

$$\begin{cases} F_{t+1} - \varphi_{t+1}^{1} s_{t+1} \le \gamma_{t+1}^{1} q_t + \beta_{t+1}^{1} \\ \vdots \\ F_{t+1} - \varphi_{t+1}^{L} s_{t+1} \le \gamma_{t+1}^{L} q_t + \beta_{t+1}^{L} \end{cases} \qquad (13)$$

[23] A key step when developing a SDDP model is estimating the cuts' parameters φ_{t+1}^{l}, β_{t+1}^{l} and γ_{t+1}^{l}. Recall that these parameters are determined during the backward optimization phase and that they are then used in the forward simulation phase to check the quality of the approximation. The procedure used to determine the cuts' parameters is detailed next using the same notation as in the work by *Tilmant and Kelman* [2007]. The objective of describing this procedure is also to provide the basis for the extended SDDP model that will be presented in the next section.

[24] To determine the values of φ_{t+1}^{l}, β_{t+1}^{l} and γ_{t+1}^{l}, one must imagine that at stage $t + 1$ the pair $(s_{t+1}^{\circ}, q_t^{\circ})$ is sampled. From that pair, K vectors of inflows q_{t+1}^{k} are generated by the following autoregressive model of order one:

$$q_{t+1} = \mu_{t+1} + \rho_{t+1,t} \frac{\sigma_{t+1}}{\sigma_t} \left(q_t^{\circ} - \mu_t\right) + \xi_t \sigma_t \sqrt{1 - \rho_{t+1,t}^2} \qquad (14)$$

where ρ, μ and σ are the estimated lag-one autocorrelation, mean and standard deviation associated with the inflows to the reservoirs. Note that residuals ξ_t are cross correlated. The water balance constraint can then be calculated

$$s_{t+2} - C_R(r_{t+1} + l_{t+1}) = s_{t+1}^{\circ} + q_{t+1}^{k} - e_{t+1}(s_{t+1}^{\circ}) - i_{t+1} \qquad (15)$$

and the corresponding dual information is $\lambda_{w,t+1}^{l,k}$, which is a $(J \times 1)$ vector, where J is the number of reservoirs. At that stage $t + 1$, the L cuts which constitute the upper bounds to the true future benefits function F_{t+2} are also constraints

$$F_{t+2} - \varphi_{t+2}^{l} s_{t+2} \le \gamma_{t+2}^{l} q_{t+1} + \beta_{t+2}^{l}, \quad l \in [1, \ldots, L] \qquad (16)$$

and the dual information is the $(L \times 1)$ vector $\lambda_{c,t+1}^{l,k}$. The dual information of the optimization problem at stage $t + 1$ can be used to derive the vector of slopes φ_{t+1}^{l} with respect

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to the storage state variable s_{t+1} to approximate the future benefit function F_{t+1} at stage t:

$$\frac{\partial F_{t+1}^k}{\partial s_{t+1}} = \lambda_{w,t+1}^{l,k} \tag{17}$$

Taking the expectation over the K artificially generated flows, we get for the jth element of the slope vector φ_{t+1}^l

$$\varphi_{t+1}^l(j) = \frac{1}{K}\sum_{k=1}^{K} \lambda_{w,t+1}^{l,k}(j) \tag{18}$$

Similarly, the jth element of the vector of slopes γ_{t+1}^l with respect to the hydrologic variable q_{t+1} can be obtained from the dual information associated with the constraints (15) and (16), i.e., from the vectors $\lambda_{w,t+1}^{l,k}$ and $\lambda_{c,t+1}^{l,k}$, respectively:

$$\begin{aligned}\frac{\partial F_{t+1}^k}{\partial q_t} &= \frac{\partial F_{t+1}^k}{\partial q_{t+1}}\frac{\partial q_{t+1}}{\partial q_t} \\ &= \left(\lambda_{w,t+1}^{l,k}(j) + \sum_{l=1}^{L}\lambda_{c,t+1}^{l,k}\gamma_{t+2}^l(j)\right) \times \left(\frac{\sigma_{t+1}(j)}{\sigma_t(j)}\rho_{t,t+1}(j)\right) \\ &= \gamma_{t+1}^{l,k}(j)\end{aligned} \tag{19}$$

The expected slope with respect to the inflows can then be determined:

$$\gamma_{t+1}^l(j) = \frac{1}{K}\sum_{k=1}^{K}\gamma_{t+1}^{l,k}(j) \tag{20}$$

The jth element of the vector constant terms is given by:

$$\beta_{t+1}^l(j) = \frac{1}{K}\sum_{k=1}^{K} F_{t+1}^k - \varphi_{t+1}^l(j)s_{t+1}^\circ(j) - \gamma_{t+1}^l(j)q_t^\circ(j) \tag{21}$$

[25] The objective function of the SDDP model (8) does not include the economic benefits associated with irrigation. In fact, the previous model solves the multiobjective (hydropower-irrigation) problem using the constraint method: one of the objectives, here irrigation, was taken out of the objective function and considered as additional inequality constraints with lower bounds corresponding to minimum irrigation demands. The Lagrange multipliers λ_w associated with the constraints (15) therefore correspond to the marginal water values reflecting hydropower only, i.e., the saved costs due to avoided thermal production. The next section describes an extension of the SDDP model that simultaneously maximizes the net benefits from both hydropower generation and irrigation, therefore providing optimal allocation decisions (reservoir releases and withdrawals) and marginal water values while still considering temporally and spatially correlated hydrologic inflows.

5. SDDP Model With Irrigation Benefits

[26] The net economic benefits from the agricultural sector are estimated from demand functions developed for each irrigation demand site. *Gibbons* [1986], *Young* [2005], and *Ward and Michelsen* [2002] review methods for assessing farmer's demand for irrigation water. In the work by *Booker and Young* [1994] marginal benefits functions for the Colorado River systems are obtained from a linear programming model that seeks to maximize irrigator profit for various water supplies and salt discharges. Regression analyses are then used to fit second-order polynomials to the data (the net benefits and the corresponding water withdrawals). In the work by *Rosegrant et al.* [2000], the profit from agricultural demand sites relies on the residual method and on a nonlinear empirical relationship between crop yield and water application. Quadratic benefit functions are fitted to the results of income-maximizing farm behavior models for various water supplies in the work by *Ward and Pulido-Velazquez* [2008].

[27] The net benefit from the agricultural sector, denoted $f_{t_f}^i$, is the sum of the benefits obtained at each irrigation demand site d as a function of the volume of water that has been delivered to the crops at that site during the irrigation season $y_{t_f}^d$. Note that these site-specific net benefit functions $f_{t_f}^{i,d}(y_{t_f}^d)$ must be linear or at least approximated by piecewise linear functions $\hat{f}_{t_f}^{i,d}(y_{t_f}^d)$ in order to be compatible with the Benders decomposition scheme in SDDP:

$$f_{t_f}^i(y_{t_f}) = \sum_d \hat{f}_{t_f}^{i,d}(y_{t_f}^d) \tag{22}$$

[28] Besides having its own net benefit function, an irrigation demand site also has its own topology in terms of return flows, i.e., the possibility that losses may eventually drain back somewhere downstream, in a reservoir different from the source. Each irrigation demand site, i.e., reservoir of accumulated water, is therefore characterized by a net benefit function, a topology, an efficiency, and a percentage of return flows. Denote $\alpha(i)$ as the percentage of irrigation losses for the irrigation demand site i that will drain back to the river system. Let C_R be the connectivity matrix of the reservoir system and C_I be the connectivity matrix of the irrigation system, i.e., $C_I(j, i) = \alpha$ when reservoir j receives return flows from the irrigation site i and/or $C_I(j, i) = -1$ when water is diverted from reservoir j to the irrigation site i.

[29] With the new state variable y_t and the above definitions, the objective function of the one stage SDDP optimization problem becomes

$$F_t(s_t, q_{t-1}, y_t) = \max\{f_t(s_t, q_t, r_t, y_t) + F_{t+1}\} \tag{23}$$

subject to the new water balance constraint which takes into account the topology of irrigation return flows:

$$s_{t+1} - C_R(r_t + l_t) - C_I(i_t) = s_t + q_t - e_t(s_t) \tag{24}$$

$$\underline{s}_{t+1} \le s_{t+1} \le \overline{s}_{t+1} \tag{25}$$

$$\underline{r}_t \le r_t \le \overline{r}_t \tag{26}$$

The next constraints provide lower and upper bounds on the monthly volumes of water diverted from the reservoirs to the irrigation areas. These constraints can have various

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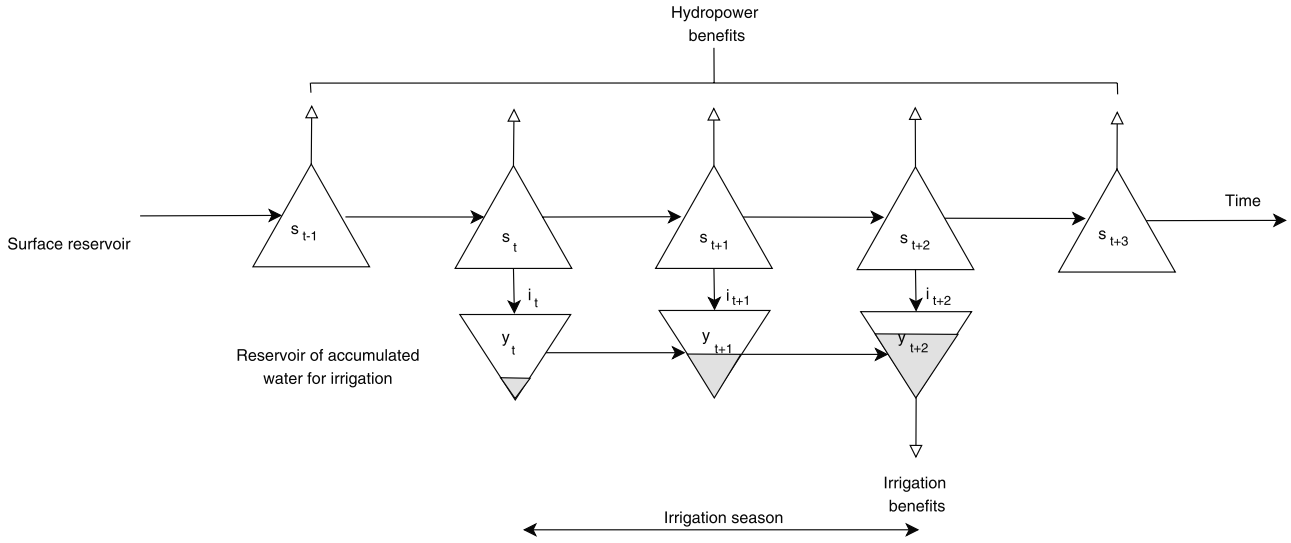


Figure 3. Reservoirs of accumulated water for irrigation.

meanings such as minimum quotas allocated to the agricultural sector, maximum pumping capacity and/or conveyance capacity, etc.

$$\underline{i}_t \le i_t \le \bar{i}_t \tag{27}$$

Lower and upper bounds on the accumulated volume of water diverted to the irrigation areas must also be specified:

$$\underline{y}_{t+1} \le y_{t+1} \le \bar{y}_{t+1} \tag{28}$$

Equation (29) updates the water balance of the "dummy" reservoir y_t at stage t (Figure 3). We can see that y_t is the net accumulated volume of water since irrigation losses are subtracted. In other words, y_t corresponds to the accumulated irrigation water requirement from the beginning of the irrigation season until the current stage t.

$$y_{t+1} - \epsilon i_t = y_t \tag{29}$$

where ϵ is the vector of irrigation efficiencies. The immediate benefit function $f_t(.)$ can now include up to three terms: (1) net benefits from hydropower generation, (2) penalties for violating operating constraints, and (3) revenues from agricultural products. Note that the third term can only be observed at the end of the growing season, when agricultural products are harvested and sold, i.e., at stage t_f. During the rest of the year, the immediate benefit function $f_t(.)$ has only the first two terms: the revenues from the production of hydroelectricity plus the penalties for not meeting operating constraints:

$$f_t(s_t, q_t, r_t, y_t) = \begin{cases} \tau_t \sum_j (\pi_t^h - \theta_t^h) c^h(j) r_t(j) - \xi_t^{\cdot} x_t & \text{if} \quad t \ne t_f \\ \tau_t \sum_j (\pi_t^h - \theta_t^h) c^h(j) r_t(j) & \\ \quad + \sum_d \hat{f}_{t_f}^{i,d}(y_{t_f}^d) & \\ \quad - \xi_t^{\cdot} x_t & \text{if} \quad t = t_f \end{cases} \tag{30}$$

Finally, the new Benders cuts approximating F_{t+1} have an additional variable, the vector y_{t+1}, and the associated vector of slopes is denoted η_{t+1}:

$$\begin{cases} F_{t+1} - \varphi_{t+1}^1 s_{t+1} - \eta_{t+1}^1 y_{t+1} \le \gamma_{t+1}^1 q_t + \beta_{t+1}^1 \\ \vdots \\ F_{t+1} - \varphi_{t+1}^L s_{t+1} - \eta_{t+1}^L y_{t+1} \le \gamma_{t+1}^L q_t + \beta_{t+1}^L \end{cases} \tag{31}$$

[30] As for the other gradients, η_{t+1}^l is also derived at stage $t + 1$ from the Lagrange multipliers $\lambda_{y,t+1}$ associated with the constraints (29):

$$\frac{\partial F_{t+1}^k}{\partial y_{t+1}} = \lambda_{y,t+1}^{l,k}(j) \tag{32}$$

Then, the expected slope of the lth cut with respect to y_{t+1} can be calculated

$$\eta_{t+1}^l(j) = \frac{1}{K} \sum_{k=1}^{K} \lambda_{y,t+1}^{l,k}(j) \tag{33}$$

Note that jth element of the vector constant terms becomes:

$$\begin{aligned} \beta_{t+1}^l(j) = & \frac{1}{K} \sum_{k=1}^{K} F_{t+1}^k - \varphi_{t+1}^l(j) s_{t+1}^{\circ}(j) \\ & - \gamma_{t+1}^l(j) q_t^{\circ}(j) - \eta_{t+1}^l(j) y_{t+1}^{\circ}(j) \end{aligned} \tag{34}$$

where y°_{t+1} is the vector of sampled accumulated water for irrigation.

[31] As mentioned early the SDDP algorithm is organized around two phases. A backward optimization phase generates the Benders cuts whose approximation power must then be evaluated. This evaluation is carried out at the end of a forward simulation phase. Because the set of Benders cuts (31) provides only an approximation to the multidimensional benefit-to-go functions, simulating the system will give a lower bound $\underline{Z}$ to the solution of

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Figure 4. Euphrates-Tigris river basin.

the multistage decision-making problem. Let M be the number of hydrologic sequences used in simulation, the expected lower bound on the optimal solution is given by:

$$\mu_{\underline{Z}} = \frac{1}{M}\sum_{t=1}^{T} f_t^m(s_t, q_t^m, r_t, y_t) = \frac{Z^m}{M} \tag{35}$$

where $f_t^m(.)$ is the immediate benefit at stage t for the hydrologic sequence $m \in [1, 2,\ldots,M]$. The standard deviation of the estimated lower bound can also be calculated

$$\sigma_{\underline{Z}} = \sqrt{\frac{1}{M-1}\sum_{m=1}^{M}\left(Z^m - \mu_{\underline{Z}}\right)^2} \tag{36}$$

The 95% confidence interval around the estimated value of $\underline{Z}$ is given by

$$\left[\mu_{\underline{Z}} - 1.96\frac{\sigma_{\underline{Z}}}{\sqrt{M}}, \mu_{\underline{Z}} + 1.96\frac{\sigma_{\underline{Z}}}{\sqrt{M}}\right] \tag{37}$$

On the other hand, at the end of the backward optimization phase, i.e., at stage one, the function $F_1(s_1,q_0,y_1)$ overestimates the benefits of system operation over the planning period when the sampled values for the state variables are s°_1, q°_0 and y°_1:

$$\overline{Z} = F_1(s_1^\circ, q_0^\circ, y_1^\circ) \tag{38}$$

If $\overline{Z}$ is inside the confidence interval, then the approximation is statistically acceptable and the problem (1) is solved. Otherwise, a new iteration is needed: a new backward recursion is implemented and the natural candidates for the sampled values of s_t and y_t are the storage volumes and accumulated volumes the previous simulations pass through. This backward phase is then followed by an new forward simulation which will exploit the cuts that have been generated during the previous backward recursions.

[32] The SDDP model described in this section is coded in MATLAB and relies on the solver LPSOLVE to solve the one stage optimization problems (23)–(31). The next two sections describe the implementation of the SDDP model (23)–(31) on the cascade of multipurpose reservoirs in the Euphrates river.

6. Euphrates River in Turkey and Syria

[33] The headwaters of the Euphrates river are located in the Taurus mountains in Turkey where 80% of the Euphrates flows are generated. In that region, the Euphrates flows southwest through steep canyons and gorges before reaching the Syrian steppe and then the Iraqi desert. The last significant tributary is the Khabur river which joins the Euphrates in Syria. Downstream, through its whole length, the Euphrates receives no further water flow. The Euphrates then merges with the Tigris river near Basra (Iraq) to form the Shatt al-Arab marshes, which empty into the Persian Gulf (Figure 4).

[34] The Euphrates has witnessed the development of irrigated agriculture by one of the first civilization (the Sumer). With the partition of the Ottoman Empire at the end of the first world war, the Euphrates became an international river shared between Turkey, Syria and Iraq. The Euphrates is a key water resource for these three countries. Irrigated agriculture has been practiced since 6000 BC and is still being developed in each country. More recently, Euphrates flows have been used to generate hydroelectricity and several dams have been constructed during the second half of the 20th century. Unfortunately, like irrigation, the hydropower potential is not homogeneous: 80% is to be found in Turkey while the remaining 20% are more or less equally distributed over Syria and Iraq.

[35] Most of the developments in the Turkish and Syrian reaches of the Euphrates have taken place over the last four decades with the Southeastern Anatolia Project (GAP) in Turkey and the Tabqa scheme in Syria. The GAP is one of the largest water resources development project in the world involving the construction of 22 dams, 19 hydroelectric power plants with an installed capacity of 7,526 MW, and the irrigation of 1.7×10^6 ha in the Turkish part of the Euphrates and Tigris river basins. The GAP has always been considered as a strategic project for Turkish authorities since the 1970s and the successive governments never gave it up despite the financial difficulties [*Kolars and Mitchell*, 1994]. In Syria, the Tabqa scheme which includes a hydropower plant (880 MW) and several irrigation districts was also planned in the 1970s. In the first project made by the Soviets 850,000 ha were to be irrigated with the water of the Tabqa reservoir (lake Assad). This rather optimistic estimate has been downgraded several times and today only 200,000 ha are operational on the left and right banks of the Euphrates [*Alia*, 2007].

[36] The development of these projects on the Euphrates have suffered from the absence of coordination between the two riparian countries. Political relations between Syria and Turkey have been characterized by mistrust, and sharing the Euphrates is just one disagreement among the many between the two countries. In this study however we will

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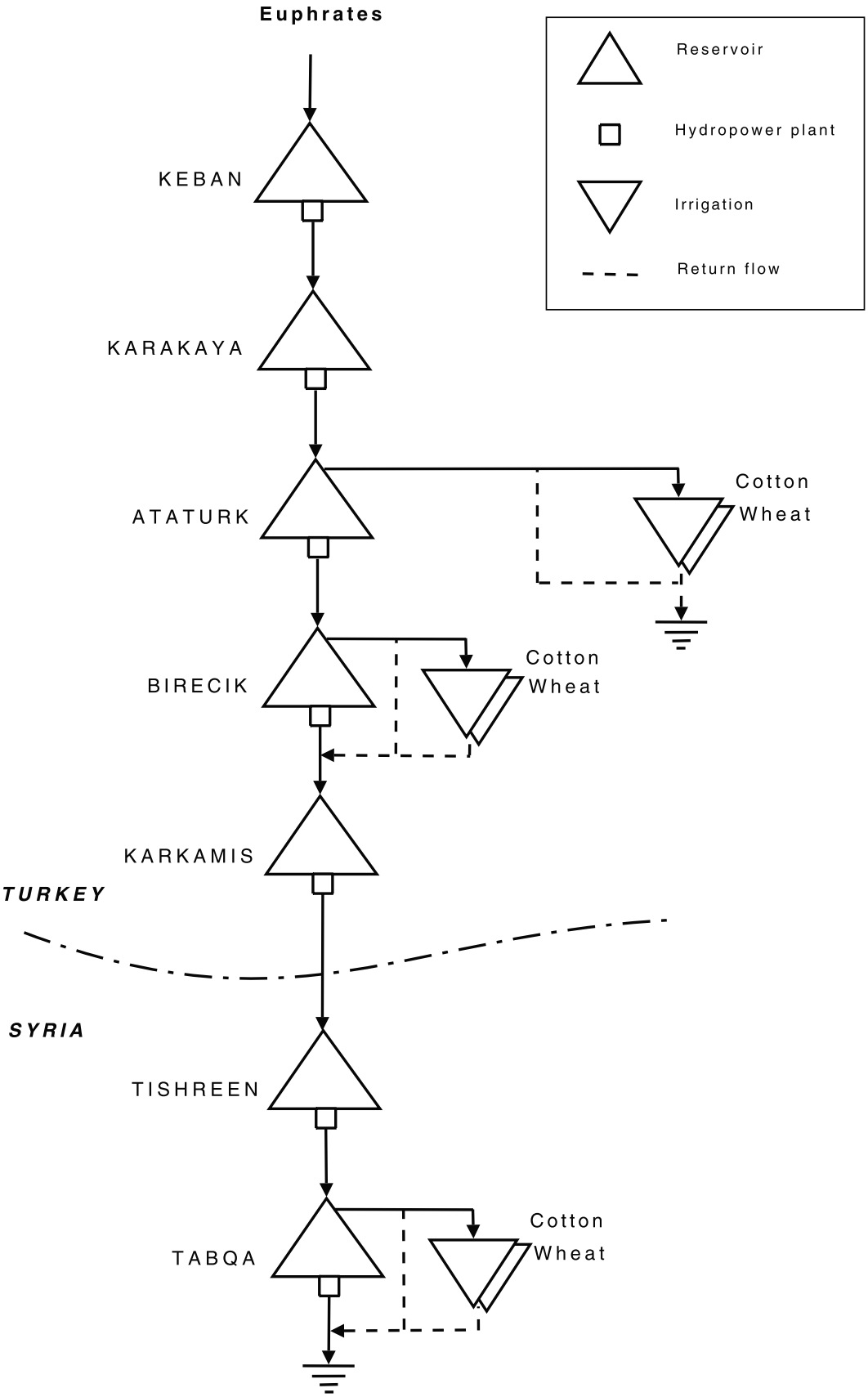


Figure 5. Euphrates system in Turkey and Syria.

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Table 1. Majors GAP Dams Considered in the SDDP Model

Name	ID	Capacity	Storage Capacity	Irrigation[a]
Keban	Kn	1240 MW	31.0 km^3	
Karakaya	Ka	1800 MW	9.58 km^3	
Ataturk	Ak	2400 MW	48.7 km^3	X
Birecik	Bk	672 MW	1.22 km^3	X
Karkamis	Ks	180 MW	0.157 km^3	
Tishreen	Tn	630 MW	1.88 km^3	
Tabqa	Ta	880 MW	14.16 km^3	X

[a]Crosses indicate when a reservoir is used for irrigation.

assume that the two riparians coordinate the allocation policies so that basin-wide benefits are maximized. Although this assumption does not reflect the current situation, it will allow us to assess the benefits forgone because of a lack of coordination when operating the system, which is the ultimate goal of a broader research initiative on the efficient sharing of water and benefits in international basins.

[37] The hydrological regime of the Euphrates at the border between Turkey and Syria is characterized by two seasons: the high-flow season slowly starting in December followed by a low-flow season starting in July. Spring snowmelt from April to June in the Taurus mountains in Turkey contributes to about 70% of the annual flow. Hence, reservoirs are expected to refill quickly during the spring and reach a maximum storage capacity around July, and then being slowly drowned down until the next spring snowmelt.

7. SDDP Model for the Euphrates River

[38] The SDDP model of the Euphrates river, depicted on Figure 5, includes 13 reservoirs: 7 storage reservoirs, 3 reservoirs of accumulated water for wheat and 3 other reservoirs of accumulated water for cotton. Table 1 lists the main characteristics of the dams which were considered in this study. As we can see from Figure 5, the main diversions for irrigation water are the Ataturk, Birecik and Tabqa reservoirs. Together these diversions will be used to irrigate more 1 million ha of mainly cereals and cotton. With a gross irrigation water demand of 14,000 m^3/ha/a, this corresponds to a volume of 14.5 km^3, which is about 50% of the natural flow at Hit in Iraq [*Altinbilek*, 2004]. The city of Aleppo in Syria also relies to a large extent on the Euphrates water. However, the impact on the availability of water in the system is minimal as only 200 hm^3/a are diverted from the Tabqa reservoir, representing less than 1% of the annual flow upstream Tabqa [*Alia*, 2007].

[39] The main parameters required to calculate the net short-run returns to irrigation water are listed in Table 2. Various sources were used but they rarely match. Details on the Tabqa scheme are taken from [*Alia*, 2007]. Background information on GAP comes [*Kolars and Mitchell*, 1994], while crop prices, yields and crop water requirements are derived from FAO databases (FAOSTAT, CLIMWAT) and from the following references [*Unlu et al.*, 2005] and [*Yilmaz et al.*, 2005]. Note that prices used here correspond to those that were applicable in year 2003 (1 US$ = 1.49 TRY (new Turkish Lira) = 46.6 SYP (Syrian Pounds)). Assuming that one ton of seed cotton will give 400 kg of fibers and 600 kg of cottonseeds, the weighted net benefits for cotton is US$800/ton and the weighted yield is 2.4 ton/ha. We further assume that 30% of the water withdrawn will drain back to the river system as suggested by *Beaumont* [1996] and by *Kolars and Mitchell* [1994]. In Figure 5 we can see that none of the irrigation return flows from Ataturk reenter the system considered in this study as they will drain back in the Khabur river, which merges with the Euphrates downstream of the Tabqa dam.

[40] Cotton is by far the dominant crop during the irrigation season in that region and both the Tabqa scheme and the GAP were planned with cotton as the main crop. In the Harran plain, one of the largest irrigation area in the Euphrates in Turkey, Cotton has been and still is the dominant crop occupying more than 90% of the cultivated area [*Ozdogan et al.*, 2006]. However, further expanding the irrigation areas in Turkey with Cotton is likely to have a major impact on the availability of water for downstream riparians, and seems therefore politically difficult as Turkey is still negotiating its membership in the EU [*Tilmant et al.*, 2007]. As a matter of fact, with the irrigation component of the GAP well beyond schedule, it is unlikely that the 250.000 ha that remain to be irrigated will be turned into cotton fields during the summer; less water demanding crops are rather to be cultivated if developments go on. These constraints on land area are taken into account to derive the net benefits from the three irrigation demand sites: Ataturk, Birecik and Tabqa. More specifically, we assume that cotton is the main crop on existing irrigated lands (380.000 ha), and that wheat will be the main crop in the irrigation schemes that are currently in an advanced planning phase or in construction, i.e., about 250.000 ha in Turkey.

[41] To assess the impact of variation in supplies on crop yields, the linear relationship between crop yield deficit and the actual evapotranspiration (ETa) is used [*Doorenbos and Kassam*, 1979]:

$$c_a^i(p) = c_{\max}^i(p)\left[1 - K_y(p)\left(1 - \frac{ET_a(p)}{ET_{\max}(p)}\right)\right] \qquad (39)$$

where $c_a^i(p)$ and $c_{\max}^i(p)$ are the actual and maximum yield of crop p, respectively, ET_a and $ET_{\max}$ are the actual and maximum evapotranspiration, respectively, $K_y(p)$ is the crop yield coefficient specific to crop p. Generally speaking, a higher K_y means a higher yield reduction for the same water deficit. We will further assume that $K_y(p)$ remains the same

Table 2. Main Irrigation Parameters: Turkey

Crop	Net Return (US$/ton)	Farm Gate Price (US$/ton)	Variable Cost (US$/ha)	Yield (ton/ha)	ε[a](%)	α (%)	t_f	ET_{max} [a](mm)
Cotton	800	1300	1205	2.4	45 (40)	30	Nov	554 (574)
Wheat	230	650	1053	2.5	45 (40)	30	Jul	300 (344)

[a]Values in parentheses are for Syria.

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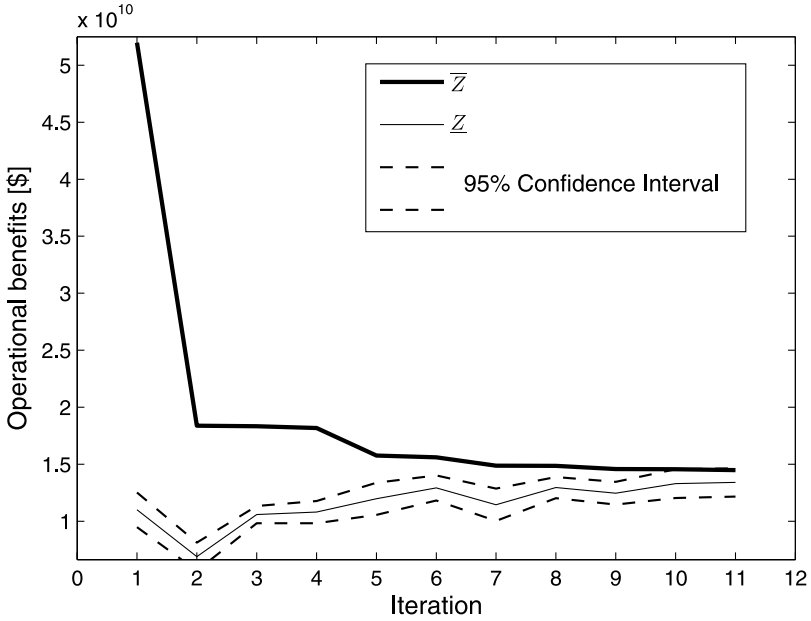


Figure 6. Convergence of the SDDP model.

in each of the stage t within the irrigation season $[t_i, t_f]$ [*Vedula and Kumar*, 1996; *Vedula and Mujumdar*, 1992]. Recall that a new state variable y_t representing the accumulated volume of water diverted to the crops from the beginning of the irrigation season until stage t is added to the state space vector. With the irrigation demand sites supplied from large reservoirs, we can assume that rationing, when it occurs, is evenly distributed over the irrigation season so that (39) can be rewritten:

$$c_a^i(p) = c_{\max}^i(p)\left[1 - K_y(p)\left(1 - \frac{y_{t_f}^{p,d}}{\bar{y}_{t_f}^{p,d}}\right)\right] \tag{40}$$

where $y_{tf}^{p,d}$ is the total volume of water supplied to crop p at site d during the irrigation season, from stage $t = t_i$ to $t = t_f$. The total crop water requirement for crop p at site d is given by $\bar{y}_{tf}^{p,d}$. The net benefit function for crop p at the irrigation demand site d can then be derived from the actual yield $c_a^i(p, d)$ [*ton*/*ha*], the farm gate price $\pi^i(p, d)$ [$/*ton*], the variable costs $\theta^i(p, d)$ [$/*ha*] and cultivated area $A(p, d)$ [*ha*].

$$f_{t_f}^{i,p,d}(y_{t_f}^{p,d}) = \left[\pi^i(p,d)c_a^i(p,d) - \theta^i(p,d)\right]A(p,d) \tag{41}$$

[42] Summing up over all irrigation demand sites (Ataturk, Birecik, Tabqa) and crops (Cotton, Wheat), the net benefit function of the agricultural sector becomes:

$$f_{t_f}^i(y_{t_f}) = \sum_{d=1}^{3}\sum_{p=1}^{2} f_{t_f}^{i,p,d}(y_{t_f}^{p,d}) \tag{42}$$

[43] For the hydropower sector, we assume that energy prices π_t^h in Turkey correspond to the short-run marginal costs (SRMC) of the Turkish hydrothermal electrical system as described by *Tilmant and Kelman* [2007]. As a matter of fact, since gas-fired power plants are likely to become the marginal power plants in Turkey, the SRMC is likely to oscillate around the variable cost of such power plants, i.e., US$40/MWh. In Syria, the value of hydropower is more expensive since it will displace more costly oil-fired power plants whose variable costs exceed US$55/MWh.

[44] The time step is 1 month and the planning horizon corresponds to 5 years ($T = 60$). The numbers of backward openings and simulation sequences are equal and set to 20 ($K = M = 20$). With these parameters, the SDDP algorithm converges after 11 iterations as illustrated on Figure 6.

8. Analysis of Optimization and Simulation Results

[45] Simulations results typically consist of allocation decisions, i.e., r_t and i_t, for each triple stage-reservoir-inflow. In this paper, we will focus on the marginal water values which are also available at the optimal solution for each triple. Recall that the marginal water values are the Lagrange multipliers associated with the mass balance equation of the reservoirs. They give the change in the system's operational benefits due to a small change in the availability of water. More precisely, we calculate the short-run water values for hydropower and irrigation since capital costs are ignored and considered as sunk costs, which is a reasonable assumption for midterm water allocation among sectors, i.e., when the capacity of the hydraulic infrastructures remains the same [*Young*, 2005].

[46] Figure 7a displays the average monthly water values for each reservoir listed in Table 1. The examination of Figure 7a reveals that the water values are decreasing as water is flowing downstream. The reason is to be found in that there is a cascade of hydropower plants with an upstream reservoir (Keban) whose outflow can further be used downstream, either for hydropower generation or for irrigation. In other words, any additional cubic meter of water in Keban is more valuable than a cubic meter of water

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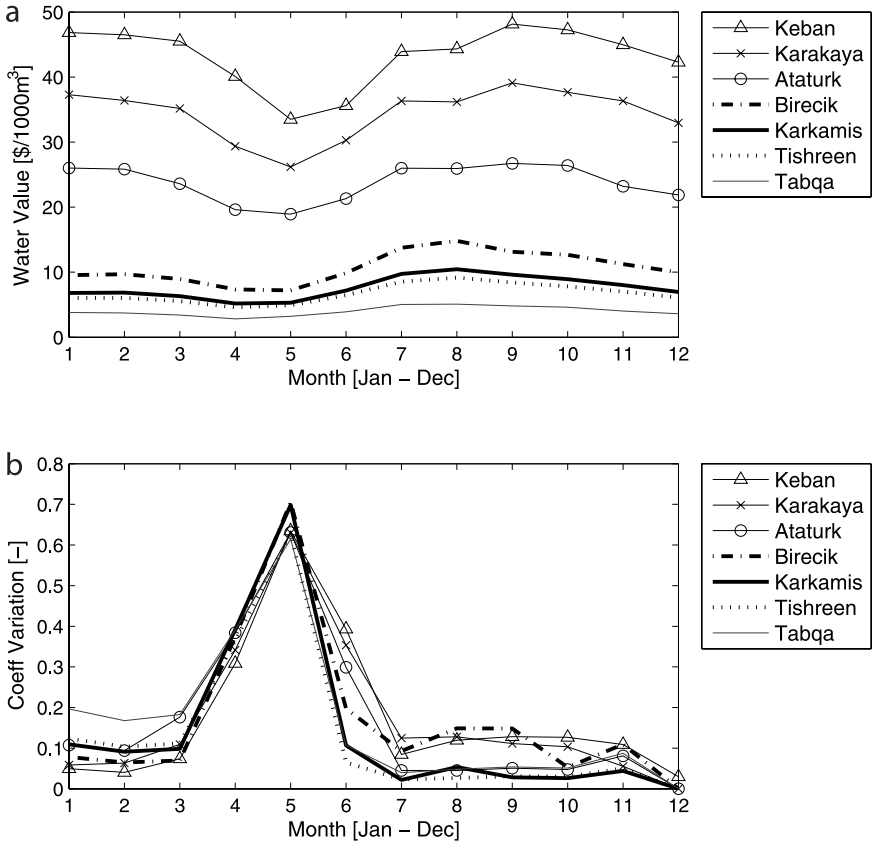


Figure 7. Average (a) water values and (b) coefficient of variation.

downstream, say at Birecik, because it can be used several times for nonconsumptive uses as it flows to the Persian Gulf. The water values at the most upstream reservoir (Keban) are more than six times higher than that estimated at Tabqa, which is the downstream site in this study. As we will see later, the Ataturk reservoir contributes to a significant portion of this difference.

[47] We can also see in Figure 7a that the water values change according to the time of the year: they are lower during the high-flow season (March–July) and vice versa. This observation is consistent with the fact that water becomes more valuable during the low-flow season when it is a scarce resource for both sectors (irrigation and energy). During the high-flow season, the water value at Keban looses about 30% of its value during the low-flow season.

[48] Of interest is how sensitive are the water values to the hydrologic conditions, and how such sensitivity changes according to the time of year. Figure 7b shows the coefficients of variation of the water values for each month of the year and each reservoir. We can clearly see that the dispersion of the water values is proportional to the inflows; it reaches a peak in May during the high-flow season and then decreases during the low-flow season. This is explained by the fact that reservoirs' outflows in May are strongly dependent on the availability of water in the system and thus on the degree with which reservoirs are being refilled. During the dry season, however, the variability decreases because reservoirs are being depleted to generate energy and to irrigate. During the low-flow season natural flows are also more constant. In fact, Keban with its huge storage capacity controls the availability of water in the system by choosing to release water or not. As we move downstream, we can see that the difference between winter and summer water values is smaller. This is because a significant portion of the inflows to the downstream reservoirs is already regulated by upstream Keban, Karakaya and Ataturk. Note that the average water values and coefficients of variation are presented in tabular form in Appendix A.

[49] Since the water value at a given site is equal to its direct (in situ) value plus the indirect value generated downstream, it is interesting to make the distinction between these two components. Figure 8 displays the direct values, i.e., the value generated at a given site. As expected from the physical characteristics of the system we can see that Ataturk, which is the largest hydropower plant and irrigation area, has the largest direct values. We can also see that the direct water values at Ataturk decrease during the high-flow season and vice versa. Note that similar conclusions can be drawn for Karakaya whose hydropower plant is the second in capacity. As already observed in Figure 7a, the water values at Keban dramatically decrease during the high-flow season and can become as low as those of Tabqa whose capacity much lower than that of Keban (880 MW versus 1240 MW). This again stresses the important role played by Keban in regulating flows to the benefits of those located downstream. The extent of those benefits can be measured by the difference in the water values during the high- and low-flow seasons.

[50] As mentioned early, water values vary with the availability of water, which in turn is a stochastic variable

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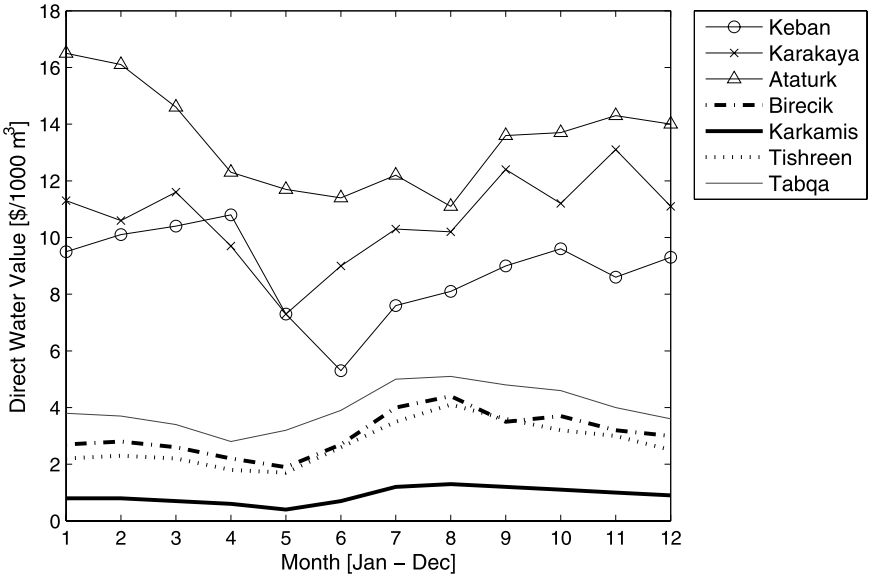


Figure 8. Direct water values.

since inflows are generated by a stochastic process. With 20 forward simulations, 20 different inflows were generated for each stage and each reservoir. Figure 9a displays the cumulative distribution functions (cdf) of water values in each reservoir in May (high-flow season), while Figure 9b shows the same functions for November, which is in the low-flow season. As expected from the examination of Figures 9a and 9b, water values are smaller and more variable during the high-flow season, especially for the three upstream reservoirs (Keban, Karakaya and Ataturk).

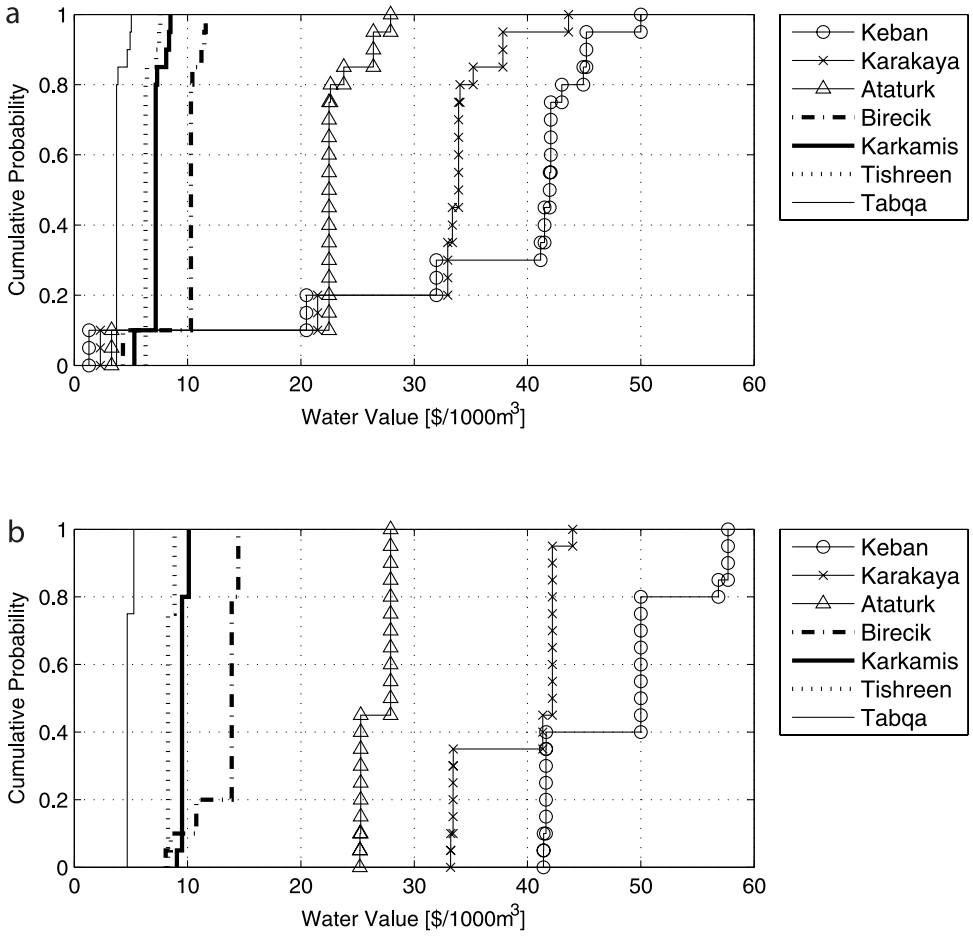


Figure 9. Cumulative distribution function of water values (a) in May and (b) in November.

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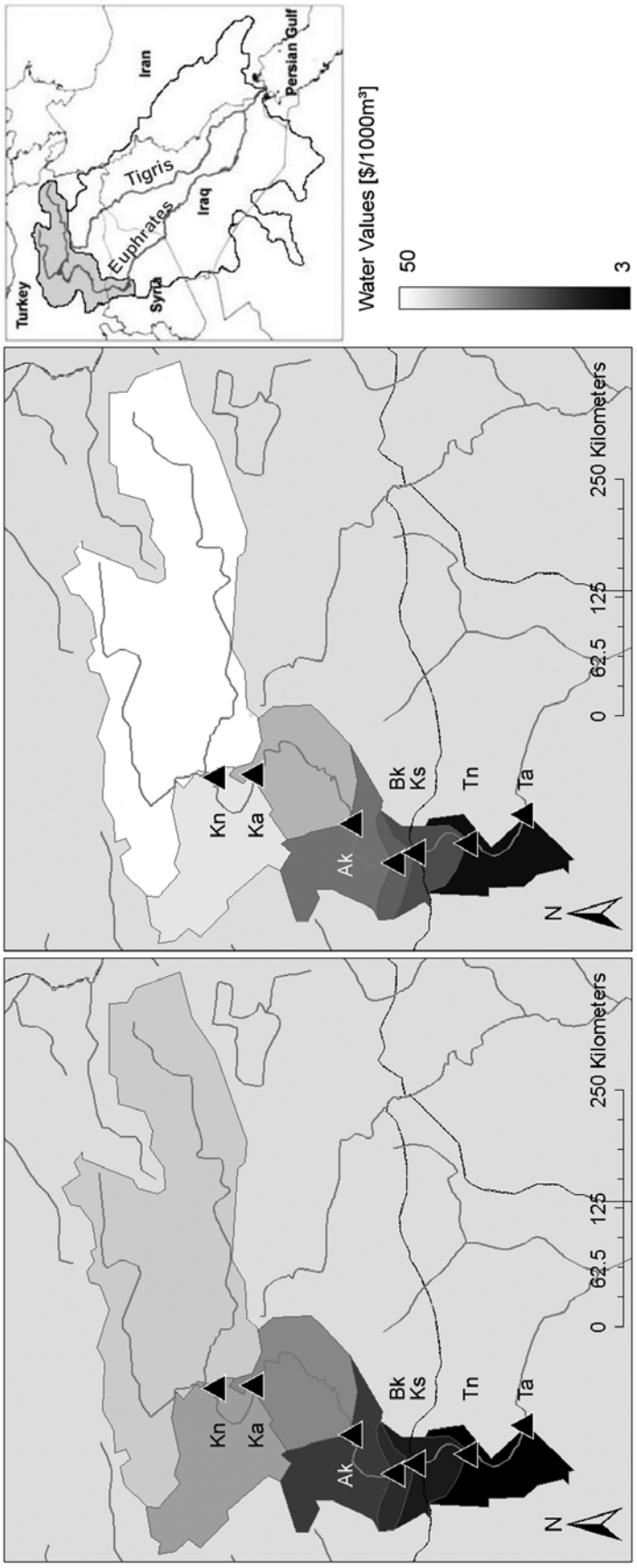


Figure 10. Map of the marginal water values in May and in November: Euphrates river basin.

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Table A1. Average Monthly Water Values[a]

Month	Keban	Karakaya	Ataturk	Birecik	Karkamis	Tishreen	Tabqa
Jan	46.8	37.3	26.0	9.5	6.8	6.0	3.8
Feb	46.5	36.4	25.8	9.7	6.9	6.1	3.7
Mar	45.5	35.2	23.6	8.9	6.3	5.6	3.4
Apr	40.1	29.3	19.6	7.3	5.2	4.6	2.8
May	33.5	26.2	18.9	7.2	5.3	4.9	3.2
Jun	35.6	30.3	21.3	9.9	7.2	6.5	3.9
Jul	43.9	36.3	26.0	13.8	9.7	8.5	5.0
Aug	44.3	36.2	25.9	14.8	10.5	9.2	5.1
Sep	48.1	39.1	26.7	13.1	9.6	8.4	4.8
Oct	47.3	37.7	26.4	12.7	8.9	7.8	4.6
Nov	45.0	36.3	23.2	11.2	8.0	7.0	4.0
Dec	42.3	33.0	21.9	10.0	7.0	6.1	3.6

[a]Unit is US$/1000 m^3.

This again illustrates the role played by the reservoirs in redistributing water throughout the year on the basis of their marginal values. In May, water values at Keban, Karakaya and Ataturk can become almost identical 20% of the time, during particularly wet years. As a matter of fact, during wet years, Keban is able to refill its reservoir in May while generating at maximum capacity in order to refill Karakaya and Ataturk. This is especially critical for the latter as its reservoir is also used to irrigate large areas in the summer. Hence, any additional cubic meter of water at Keban cannot bring further benefits compared to another cubic meter at Karakaya, especially during above-normal wet years.

[51] The marginal water values discussed so far were essentially attached to a reservoir, i.e., to a node of the system depicted on Figure 5. However, we can see in the mass balance equations of the reservoirs, i.e., equation (24), that the right-hand sides include the aggregated lateral inflows q_t which drain the intermediate basins between the reservoirs. The marginal water value has therefore a spatial dimension that corresponds to the drainage area of those intermediate basin; any additional cubic meter generated in a given intermediate basin has the same value as it will end up in the reservoir that drains this basin. In almost fully allocated and complex river basins, the mapping of marginal water values can be useful for identifying and prioritizing zones of intervention for the development and implementation of water conservation measures. Figure 10 maps the marginal water values at the beginning of the high-flow season in the Euphrates, from the headwater in Turkey to the Tabqa dam in Syria.

9. Conclusions

[52] Stochastic programming can be used as a deductive method to derive marginal water values and their statistical distributions in multipurpose multireservoir systems. By formulating the reservoir operation problem as a stochastic optimization problem, the marginal water values are the Lagrange multipliers associated with the water balance equations. They give the change in the operational net benefits due to a small perturbation in the availability of water at the different sites where a mass balance equation is computed. Since irrigation return flows are considered in the water balance equation, the water value calculated by the SDDP model has two of the five components listed by *Rogers et al.* [2002]: the marginal value of product and the benefits from return flows.

[53] The use of SDDP as a deductive method is illustrated with a cascade of reservoirs in the Turkish and Syrian parts of the Euphrates river. Downstream uses and benefits in Iraq could not be taken into account as data were not available. Hence, results are given for illustrative purposes only. If Iraq were to be included, then the water resources of the whole Euphrates-Tigris river basin could be jointly optimized since both river merge in Iraq. The average water values presented in Figure 7 would likely change as downstream water benefits and costs would be considered (including the value of the Iraqi marshes).

[54] The proposed methodology assesses the short-run marginal water values since fixed costs are ignored and considered as sunk costs. This assumption was motivated by the fact that the construction of these huge hydraulic infrastructures (dams, power plants, irrigation systems) is also driven by domestic and international objectives such as food and energy security, strengthening ties with the Kurdish minority in Turkey and thus undermining the separatist movement, boosting the economy of regions characterized by an endemic stagnation, etc. The short-run marginal water values can then be used in midterm allocation studies among sectors (here hydropower and irrigation).

[55] The analysis of simulation results for different hydrologic scenarios reveals that the water values are higher during the low-flow season and vice versa. The variance also decreases as water flows downstream and is being more and more regulated by the successive reservoirs. The direct water value, i.e., the portion of the water value generated at a given site, varies significantly from one reservoir to the other. In fact, the direct water values are affected by both technical and economical factors such as the hydropower plant capacity and production coefficient, the irrigation area, efficiency and productivity. From the simulation results it is also clear that upstream reservoirs play an important role in regulating flows downstream to the benefits of downstream users.

Appendix A

[56] Table A1 lists the average water values for each reservoir and each month. They give the change in the system's operational benefits due to a small change in the availability of water in each reservoir.

Table A2. Coefficient of Variation: Monthly Water Values[a]

Month	Keban	Karakaya	Ataturk	Birecik	Karkamis	Tishreen	Tabqa
Jan	0.0493	0.0586	0.1073	0.0782	0.1094	0.1235	0.1963
Feb	0.0404	0.0633	0.0941	0.0646	0.0912	0.1034	0.1676
Mar	0.0738	0.1095	0.1764	0.0695	0.0984	0.1116	0.1825
Apr	0.3086	0.3414	0.3839	0.3739	0.3919	0.3656	0.3917
May	0.6348	0.6298	0.6309	0.7053	0.6983	0.6272	0.6145
Jun	0.393	0.3533	0.2989	0.1978	0.1067	0.065	0.1075
Jul	0.0845	0.1249	0.045	0.093	0.0219	0.0232	0.0394
Aug	0.1196	0.1281	0.0451	0.1486	0.0534	0.0267	0.0482
Sep	0.1281	0.1112	0.0507	0.1485	0.0282	0.0308	0.0538
Oct	0.1269	0.1034	0.0474	0.0531	0.0263	0.03	0.0509
Nov	0.1089	0.0555	0.081	0.1093	0.0438	0.0503	0.0872
Dec	0.0298	0	0	0	0	0	0

[a]Unitless.

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[57] Table A2 lists the direct water values for each reservoir and each month. They give the individual contribution of each reservoir to the total water value at any given site throughout the basin.

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